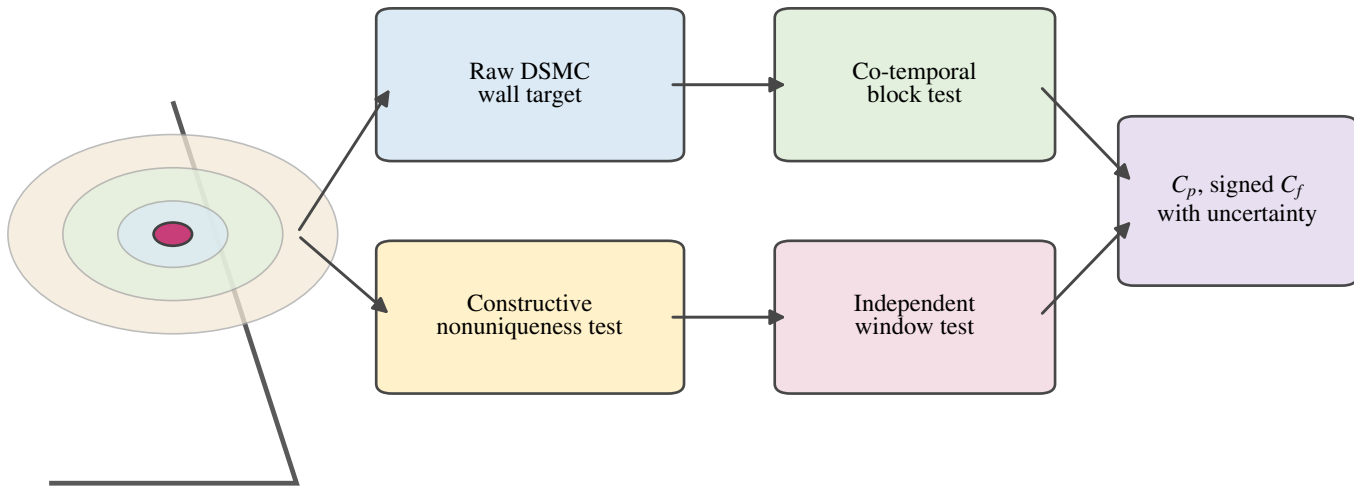


(b) Physics-first validation logic



The neural surrogate is a comparison tool; the central claim is tested with molecular collision statistics and direct DSMC surface impulses.